

Team

Capital Theta.

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FarmAssist

AI Farm Decision & Market Intelligence Agent

Grounded, explainable farming recommendations — powered by live weather, mandi prices, MSP data, and Google Gemini.

01 The Problem

Farming decisions are high-stakes and time-sensitive — but the information behind them isn't.

1 Fragmented sources

Weather, market prices, and pest advisories live on separate portals, apps, and SMS lines — no single view.

2 Generic advice

Government advisories and forums give one-size-fits-all guidance that ignores a farmer's specific soil, stage, and location.

3 Stale market data

By the time price info reaches a farmer, the mandi may have already moved — costing real income.

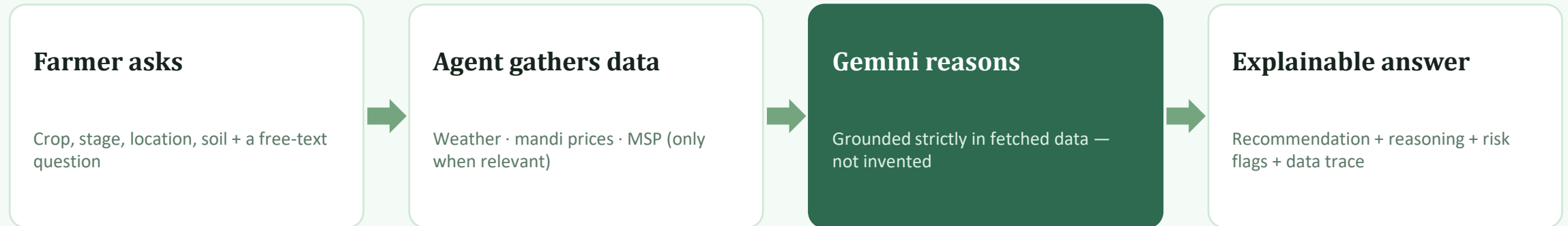
4 Wrong timing = crop loss

Irrigating, fertilizing, or selling at the wrong moment (e.g. into a heatwave or a price dip) directly hits yield and income.

02 The Solution

An AI agent that orchestrates real data sources, then reasons over them — not a chatbot guessing from memory.

FarmAssist combines live weather, market prices, and MSP data behind one conversational interface, and always shows its work.



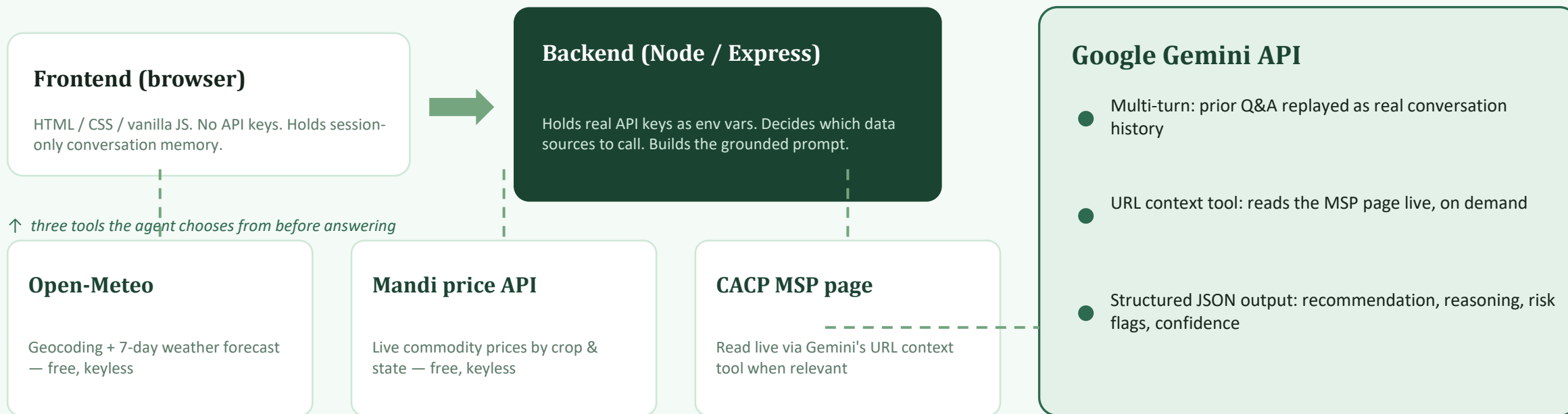
Why this matters: the LLM never fabricates a price or a forecast — it only reasons over numbers the backend actually fetched this turn.



02

Solution — Architecture

A clean client/server split keeps API keys server-side and the agent's decisions auditable.



Response: recommendation · reasoning · risk flags · confidence · full data trace — sent back to the farmer, with nothing hidden.

03

User Interface

A single, farmer-facing form — plain language in, a grounded, explainable answer out.



Simple input

Crop, growth stage, location, soil type, and a free-text question — no jargon, no menus to learn.



Conversation memory

Follow-up questions build on earlier turns in the same session, shown with a “remembering N turns” badge.



Explainability panel

Every answer is paired with the exact weather, mandi and MSP data behind it — no black box.



Responsive & lightweight

Single-page app, works on a low-end phone browser or desktop, no install required.

Languages & tech used

HTML5

CSS3

JavaScript

Node.js

Express

REST APIs

04

AI Model — Why Gemini?

Chosen for capabilities this exact problem needs, not just familiarity.

Tool / function calling

Lets the agent decide which data source (weather, mandi, MSP) a question actually needs, instead of one rigid pipeline.

Built-in URL context tool

Reads a live webpage directly — no custom scraper needed for the MSP lookup.

Multimodal by default

Same API accepts images — a natural next step for photo-based pest/disease identification.

Fast + affordable at scale

Gemini 2.5 Flash keeps latency and cost low enough for a real-time farmer-facing tool; Pro available for harder reasoning.

04

AI Model — Fetching Mandi Prices

The API key lives on the server; the model only ever sees the numbers it returns.

How it works

- 1 Backend resolves the farmer's location to a state via geocoding
- 2 It calls a free, keyless mandi price API with { crop, state }
- 3 Live min / modal / max prices (₹/quintal) come back as structured JSON
- 4 Only those real numbers are inserted into Gemini's prompt — never invented

Example returned data

Crop: Wheat · **State:** Uttar Pradesh

Baberu APMC — modal ₹2,700/qtl

Shahganj APMC — modal ₹2,600/qtl

Akbarpur APMC — modal ₹2,567/qtl

Sikandraraau APMC — modal ₹2,212/qtl

Source: mandi-api.onrender.com (data.gov.in) — covers Maharashtra, Uttar Pradesh, Punjab, Madhya Pradesh, Karnataka.

AI Model — MSP as a Coverage Fallback

Live mandi data doesn't reach every district yet, so MSP fills the gap as a trustworthy floor price.

The limitation

The keyless mandi price API only covers five states in its current version. Outside them — or for crops it has no listing for — there is no live per-mandi price to show.

No public, real-time, nationwide input-cost or price API exists to fall back on either — so guessing a number was never an option.

The fix

Gemini's URL context tool reads the official CACP MSP page live, on demand — giving every crop a nationally consistent government reference price, regardless of mandi coverage.

Why this is the right trade-off

- MSP is a single, government-set number per crop — no state coverage gap
- Only fetched when the question is actually about selling or price
- If the page is unreachable, the agent says so plainly — never fabricates a figure
- Sets up a clean upgrade path: swap in more mandi coverage later without changing the agent's reasoning

05

Working Screenshot — Weather + Market + MSP

One question exercising pest diagnosis, irrigation timing, and a live MSP-vs-mandi price comparison.

FarmAssist

Ask about sowing, irrigation, pests or selling — grounded in live weather and market data.

Crop

Growth stage

Rice

Pre-sowing

Location (village / town / district)

Ghaziabad, Uttar Pradesh

Soil type (if known)

alluvial

Your question

e.g. Should I irrigate this week, or wait? Any pest risk given the weather?

Get recommendation

New conversation

A single detailed farmer question covering pest symptoms, irrigation, and selling.

My rice crop is in the flowering stage. Over the last 3 days I've noticed yellowish-brown patches on the lower leaves spreading upward, with leaf tips drying out, and when I part the crop near the base I can see small brown hopper-like insects clustered on the stems. I applied urea 15 days ago and haven't added anything since. Given this and the weather forecast, is this a pest or disease problem and what should I do? Also — should I irrigate this week, and is now a good time to sell my last batch of paddy at the mandi, or should I wait? How does the current mandi price compare to MSP?

Recommendation

Drain standing field water immediately to suppress the Brown Plant Hopper pest infestation, hold off on irrigation, and sell your stored paddy at higher-paying mandis where prices beat the MSP.

The insects clustering at the base and yellow-brown drying leaves indicate a Brown Plant Hopper (BPH) attack aggravated by your recent urea application at flowering stage; draining water disrupts their habitat, and withholding irrigation is advised even though 7-day weather data is currently unavailable. For your harvest, the 2026-27 Paddy MSP is Rs 2,441 per quintal (Common) and Rs 2,461 per quintal (Grade A). Because prevailing modal prices in nearby UP mandis like Baberu (Rs 2,700/qtt), Shahganj (Rs 2,600/qtt), and Akbarpur (Rs 2,567.29/qtt) are higher than MSP, it is a good time to sell, provided you avoid depressed markets like Sikandraraau (Rs 2,212.35/qtt).

Confidence: medium

Pest outbreak (BPH)

Hopperburn risk

Missing weather data

Price variation below MSP

Watch for: Heavy rainfall or rapid price changes at your nearest mandi could impede field drainage or alter the selling advantage.

The agent reads MSP live (₹2,441–2,461/qtt) and compares it against real mandi prices across four markets.

Multi-source in one answer

Combines pest diagnosis, an irrigation call, and a live MSP-vs-mandi comparison in a single response.

Named, comparable numbers

Baberu, Shahganj, Akbarpur, and Sikandraraau prices are each compared against MSP, not averaged away.

Graceful gaps

Flags “Missing weather data” honestly instead of pretending irrigation timing is certain.

05

Working Screenshot — Multi-turn Memory

Follow-up question after a pest diagnosis, answered without repeating any context.

FarmAssist
Ask about sowing, irrigation, pests or selling — grounded in live weather and market data.

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alluvial

Your question

e.g. Should I irrigate this week, or wait? Any pest risk given the weather?

The input form: crop, stage, location, soil, and question.

Get recommendation

New conversation

If I go with the treatment you suggested for the pest problem, will that affect when I can safely apply my next round of fertilizer?

Recommendation

Yes, you must hold off on applying any fertilizer until the brown planthopper infestation is completely controlled and normal moisture is restored.

Draining the field to control the hoppers leaves the root zone dry, meaning fertilizer cannot be properly absorbed, especially with dry weather ahead (only 1.5mm rainfall on day 1 and 0.0mm for the remaining 6 days). Furthermore, applying nitrogenous fertilizer like urea fuels rapid brown planthopper multiplication and worsens hopperburn, especially under warm conditions up to 34.8C and 71% average humidity. Additional top-dressing should be avoided during flowering under active pest stress until the crop has fully stabilized.

Confidence: high

Pest resurgence risk

Hopperburn risk

Nutrient burn risk

Remembering 1 earlier turn

Watch for: If hopper populations do not decline within a few days of draining, insecticide application may be required before any field re-flooding or nutrient application.

Follow-up answer — correctly ties fertilizer timing to the earlier pest treatment, and shows “Remembering 1 earlier turn.”

Grounded reasoning

Cites specific rainfall (1.5mm / 0.0mm) and humidity (71%) from live weather data.

Real memory

References the brown planthopper problem from the previous turn without being re-told.

Honest risk flags

Surfaces “Nutrient burn risk” and “Pest resurgence risk” rather than a flat yes/no.

07

Why This Is an Innovation

Not just an LLM wrapper — a system designed so the model is structurally prevented from guessing.

1

Turns static gov't data into a live API

Most official agricultural data (like MSP) has no public API. Gemini's URL context tool reads the live government page directly on every relevant query — no scraper to write, license, or maintain.

2

Structurally can't hallucinate a number

It's not just prompted to avoid guessing — the backend only ever inserts data it actually fetched that turn. If nothing was fetched, there's nothing for the model to invent.

3

Coverage-aware hybrid pricing

When live per-mandi prices aren't available for a state, the agent automatically falls back to the nationally consistent MSP — a deliberate two-tier design, not a missing feature.

4

Explainability is the response schema, not a bolt-on

Every answer is required to return its own evidence trail — recommendation, reasoning, risk flags, confidence, and data trace all travel together, by construction.

The innovation isn't "using an LLM for agriculture" — it's making that LLM provably honest about what it does and doesn't know.

Benefits & Accuracy

Designed so trust is earned by what's visible, not claimed in a sentence.

Benefits

- One interface for weather, price, MSP, and reasoning
- Session memory — natural follow-up questions
- No live-API cost floor: weather & mandi lookups are free/keyless
- Extensible: photo pest-ID and regional languages are drop-in additions

Accuracy & Trust

- Every prompt is instructed to use only fetched data — never invent a number
- A visible confidence level (high / medium / low) on every answer
- The full data trace (weather, mandi, MSP source) is shown alongside the recommendation
- Missing or unreachable data is reported plainly, not silently guessed

References & Data Sources

- **Google Gemini API** aistudio.google.com — reasoning, multi-turn chat, URL context tool
- **Open-Meteo** open-meteo.com — free geocoding & 7-day weather forecast API
- **Mandi Price API** mandi-api.onrender.com — free, keyless market prices sourced from data.gov.in
- **CACP — Minimum Support Price** cacp.da.gov.in/Home/MSP — official government MSP reference
- **data.gov.in** Open Government Data Platform India — underlying source for mandi price data

Thank you.